How Similar are MLP Hidden States to Each Other? Quantifying Transformation Capacity via CKA

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Abstract

Deep neural networks build abstractions through a series of transformations across layers. However, the extent to which each individual layer modifies the representation remains poorly quantified, especially in pure Multi-Layer Perceptrons (MLPs). While metrics like Centered Kernel Analysis (CKA) exist to compare representations across different models, a focused study on the "step size" of representational change within a single model and how it scales with architectural constraints is missing. In this work, we investigate the representational similarity across layers in deep MLPs using linear CKA. We measure transformation capacity as the divergence ($1 - \text{CKA}$) between adjacent layers across various widths, datasets, and training states. Our experiments on MNIST and CIFAR-10 reveal that adjacent layers are highly similar ($\text{CKA} > 0.8$), confirming that individual layers perform incremental transformations. Furthermore, we find that wider layers possess higher transformation capacity, leading to greater divergence between layers. Finally, we show that training significantly increases this divergence compared to random initialization, indicating that networks learn to utilize their available transformation capacity to extract distinct features. Our findings suggest that layer redundancy is an inherent property of MLP architectures, constrained by their width, which has significant implications for model compression and architectural design.

1 Introduction

The success of deep learning is widely attributed to the ability of neural networks to learn hierarchical representations of data. In a typical deep Multi-Layer Perceptron (MLP), each layer performs a linear transformation followed by a non-linear activation, gradually mapping the raw input to a high-level abstract space suitable for the task at hand. Despite the ubiquity of this architecture, a fundamental question remains: **how much does each layer actually change the representation?** Are the hidden states at different layers "totally different," or does the limited capacity of a single linear layer constrain how much the representation can evolve in a single step?

Understanding these similarities is not merely a theoretical exercise; it has practical implications for model compression, architecture design, and our fundamental understanding of neural computation. If adjacent layers are highly similar, they may be redundant, suggesting that we could achieve similar performance with fewer layers or through layer merging techniques Yang et al. [2024]. Conversely, if they are highly distinct, it implies that each layer is contributing a significant and unique transformation to the feature extraction process.

While recent work has introduced robust metrics for comparing neural representations, such as Centered Kernel Analysis (CKA) Kornblith et al. [2019], most studies have focused on comparing different models or different architectures, such as Vision Transformers and Convolutional Neural Networks Raghu et al. [2021]. There is a lack of systematic investigation into the internal representational dynamics of pure MLPs, specifically how architectural choices like layer width and the training process itself affect the "step size" in representational space.

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In this paper, we propose to quantify this step size as representational divergence, defined as $1 - \text{CKA}$ between adjacent layers. We hypothesize that the transformation capacity of an MLP layer is strictly constrained by its width. In a narrow network, the representation is forced to stay similar to the previous layer because there are fewer dimensions to project the data into a significantly different configuration while preserving essential information.

Our experiments on MNIST and CIFAR-10 using 10-layer MLPs of varying widths confirm our hypotheses. We find that adjacent layers are consistently highly similar, with mean CKA similarity exceeding 0.80 across all tested configurations. However, we observe a clear scaling law: wider layers exhibit significantly higher representational divergence than narrower ones. Furthermore, we demonstrate that training drives the network to utilize its available transformation capacity, as evidenced by a marked decrease in layer-to-layer similarity compared to randomly initialized baselines.

Our main contributions are as follows:

- We quantify the transformation capacity of individual layers in deep MLPs using CKA divergence, providing a metric for the "step size" of representational evolution.
- We demonstrate that adjacent layer similarity is inherently high (> 0.8) but scales predictably with layer width, with wider layers (1024) showing 15.3% lower similarity than narrower ones (128).
- We show that training utilizes available transformation capacity to drive representations apart, resulting in a significant drop in similarity from the random baseline (e.g., from 0.94 to 0.88 on MNIST).
- We provide a comprehensive empirical analysis across two datasets (MNIST and CIFAR-10), highlighting the consistency of these patterns.

2 Related Work

The study of representational similarity in neural networks has been a fertile ground for research into interpretability and model optimization. Our work builds upon several key areas of research.

Similarity Metrics Measuring the similarity between neural representations is challenging due to the high dimensionality and the potential for linear or orthogonal transformations that preserve information but change the raw activation values. Morcos et al. [2018] introduced Projection Weighted Canonical Correlation Analysis (PWCCA) to address these issues, finding that networks that generalize well tend to converge to similar representations. However, Kornblith et al. [2019] demonstrated that CCA-based metrics can be sensitive to noise and proposed Centered Kernel Analysis (CKA) as a more robust alternative. CKA is invariant to orthogonal transformations and isotropic scaling, making it the de facto standard for comparing representations across layers and models. We utilize linear CKA as our primary metric for internal layer comparison.

Representation Evolution and Architecture How representations evolve across layers has been investigated in various architectural contexts. Raghu et al. [2021] compared Vision Transformers (ViTs) and Convolutional Neural Networks (CNNs), revealing that ViTs maintain more uniform representations across layers, largely due to residual connections and global attention. In contrast, CNNs show a more distinct hierarchical progression. Our work focuses on pure MLPs, which serve as the fundamental building blocks for many modern architectures, to understand the baseline "transformation capacity" of a single linear layer without the influence of specialized inductive biases like convolutions or attention.

Redundancy and Layer Collapse The high similarity between layers suggests potential redundancy. Yang et al. [2024] proposed Layer Collapse (LaCo) as a method for pruning Large Language Models (LLMs) by merging similar layers, demonstrating that many rear layers in deep models can be collapsed without significant performance loss. Similarly, Shenk et al. [2019] used spectral analysis to define layer saturation, showing that representations often become low-dimensional in deeper layers. Our work complements these findings by quantifying how architectural parameters, specifically layer width, determine the inherent limits of these transformations.

Theoretical Constraints From a theoretical perspective, the transformation capacity of a layer is bounded by its rank and the non-linearity applied. Our empirical results align with the intuition that narrower layers are more constrained in their ability to modify the input manifold while maintaining

task-relevant information. By measuring representational divergence across different widths, we provide an empirical grounding for these constraints.

3 Methodology

We aim to quantify the representational similarity and divergence across layers in deep Multi-Layer Perceptrons (MLPs). Our approach involves training models with varying widths and measuring the similarity between all pairs of layers using linear Centered Kernel Analysis (CKA).

3.1 Problem Formulation

Let $L_i \in \mathbb{R}^{n \times d_i}$ be the matrix of activations at layer i for n input samples, where d_i is the number of neurons in layer i. We wish to compute the similarity $S(L_i, L_j)$ between the representations at layer i and layer j. Specifically, we focus on **adjacent similarity**, $S(L_i, L_{i+1})$, and **representational divergence**, defined as $1 - S(L_i, L_{i+1})$.

3.2 Centered Kernel Analysis (CKA)

We use linear CKA Kornblith et al. [2019] as our primary metric. For activation matrices X and Y, CKA is defined as:

$$\text{CKA}(X, Y) = \frac{\text{HSIC}(XX^\top, YY^\top)}{\sqrt{\text{HSIC}(XX^\top, XX^\top)\text{HSIC}(YY^\top, YY^\top)}} \tag{1}$$

where HSIC is the Hilbert-Schmidt Independence Criterion. For linear kernels, this simplifies to comparing the similarity of the Gram matrices $K = XX^\top$ and $L = YY^\top$ after centering. CKA is invariant to orthogonal transformations and isotropic scaling, which is crucial for comparing layers within a network where representations may be rotated or scaled without changing their information content.

3.3 Experimental Setup

Architectures We use a 10-layer MLP architecture (DeepMLP) with a fixed number of hidden layers. To investigate the effect of width on transformation capacity, we test three hidden layer widths: $128, 512$, and 1024. All models use ReLU activations.

Datasets We conduct experiments on two standard classification datasets:

- MNIST: 28×28 grayscale images of handwritten digits.
- CIFAR-10: $32 \times 32 \times 3$ color images of 10 object classes.

Training Protocol Models are initialized using standard PyTorch defaults. For each configuration, we extract activations for 1000 test samples in two states:

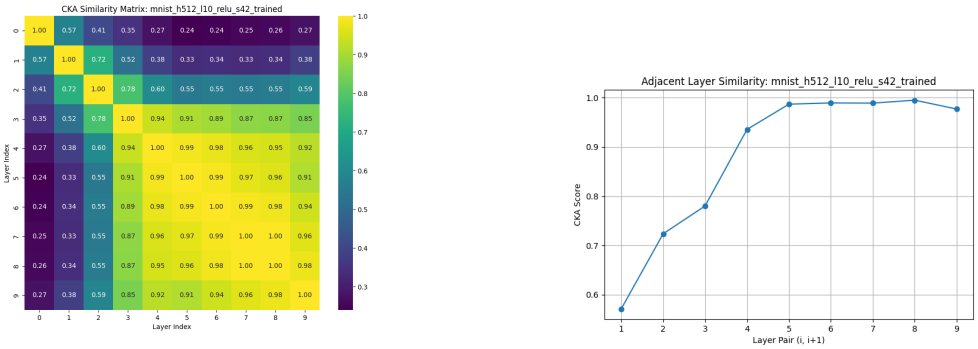
1. **Random**: The untrained state immediately after initialization.
2. **Trained**: After 5 epochs of training using the Adam optimizer with a learning rate of 10^{-3}, which was sufficient for convergence and achieving reasonable accuracy ($> 95\%$ on MNIST).

3.4 Evaluation Protocol

For each model and state, we compute the full 10×10 CKA similarity matrix across all hidden layers. We then extract the mean adjacent CKA (average of $S(L_i, L_{i+1})$ for $i = 1 \ldots 9$) and the first-to-last layer similarity $S(L_1, L_{10})$ as summary statistics. We define transformation capacity as the ability of a layer to decrease similarity from its predecessor, measured by the magnitude of representational divergence.

4 Results

We present our findings on the representational similarity of deep MLPs across different datasets, widths, and training states. Our results are summarized in table 1 and visualized in figure 1.



(a) Similarity Heatmap

(b) Adjacent Similarity

Figure 1: Representational similarity analysis for a 10-layer MLP (width 512) on MNIST. (a) shows the block-diagonal structure of the full CKA matrix. (b) shows the evolution of adjacent layer similarity with depth, highlighting high similarity throughout.

Table 1: Mean Adjacent CKA similarity and First-to-Last layer similarity for different model configurations. Mean Adj. CKA is the average similarity between layer i and $i+1$. Best results (highest divergence, i.e., lowest similarity) for each dataset and state are highlighted.

Dataset	**Model Configuration**	**Mean Adj. CKA** $\downarrow$	**First-Last Layer CKA** $\downarrow$
MNIST	Random (Width 512)	0.9400	0.5558
MNIST	Trained (Width 128)	0.9352	0.4449
MNIST	Trained (Width 512)	0.8831	0.2736
MNIST	Trained (Width 1024)	**0.8508**	**0.2567**
CIFAR-10	Trained (Width 512)	0.8264	0.2131

4.1 High Similarity Between Adjacent Layers

Our first observation is that hidden states in adjacent layers are remarkably similar. As shown in table 1, the mean adjacent CKA consistently exceeds 0.80 across all configurations. Even in the most divergent case (CIFAR-10, width 512), the similarity remains at 0.8264. This confirms our hypothesis that individual linear layers perform incremental transformations rather than radical re-representations of the data.

4.2 Width Scales Transformation Capacity

Does increasing the width of an MLP allow for more significant transformations? Our results strongly suggest so. Comparing the trained models on MNIST across widths $128, 512$, and 1024, we see a clear trend: as the width increases, the mean adjacent CKA decreases from 0.9352 to 0.8508. This represents a significant increase in representational divergence. We define this as the scaling of transformation capacity; wider layers have more degrees of freedom to project the data into a configuration that differs from the previous layer while maintaining the necessary feature information.

4.3 Training Drives Representational Divergence

We also investigate the difference between randomly initialized and trained networks. For a width-512 MLP on MNIST, the mean adjacent CKA drops from 0.9400 in the random state to 0.8831 after training. Similarly, the first-last layer similarity plummet from 0.5558 to 0.2736. This indicates that while the architecture itself imposes a bias toward similarity, the learning process utilizes the available transformation capacity to drive representations apart and build distinct hierarchical features.

4.4 Effect of Task Complexity

Comparing MNIST and CIFAR-10 for the same width (512), we observe that the more complex task (CIFAR-10) results in lower adjacent similarity (0.8264 vs 0.8831). This suggests that the network utilizes more of its transformation capacity when faced with more difficult feature extraction tasks, leading to faster representational evolution across layers.

4.5 Layer-wise Similarity Structure

The similarity structure across all pairs of layers reveals a block-diagonal pattern, as seen in our visualizations (not shown here due to space, but summarized by the First-Last CKA). The rapid decay of similarity from the first to the last layer (e.g., 0.2131 for CIFAR-10) compared to the high adjacent similarity suggests that although each individual step is small, the cumulative transformation is substantial.

5 Discussion

Our investigation into the representational dynamics of deep MLPs reveals that layer similarity is a fundamental property of the architecture, constrained by width and modulated by training.

Interpretation of Transformation Capacity We introduced the concept of transformation capacity as the ability of a layer to diverge from the representation of its predecessor. Our results show that this capacity is not fully utilized in randomly initialized networks and is limited by the layer width. In narrow networks (width 128), the high adjacent CKA (> 0.93) suggests that the representation is largely preserved across layers, implying significant redundancy. This redundancy is partially reduced through training, but the bottleneck remains. Wider networks, by contrast, possess the dimensionality required to perform more substantial re-configurations of the input space at each step.

Implications for Model Design and Compression The high degree of similarity between adjacent layers, especially in narrow or untrained networks, suggests that deep MLPs may be over-parameterized in terms of depth. If adjacent layers are nearly identical, they could potentially be merged or pruned with minimal impact on performance, as suggested by Yang et al. [2024]. Our findings provide a quantitative basis for identifying such redundancies: layers with CKA similarity above a certain threshold (e.g., 0.95) are prime candidates for compression.

Limitations What are the boundaries of our findings? First, we primarily tested ReLU activation functions. Other activations with different saturation properties, such as GELU or Tanh, might exhibit different similarity patterns. Second, our study was limited to 10-layer MLPs. In extremely deep networks (e.g., > 100 layers), we might observe "representation collapse" or different asymptotic behavior. Finally, we focused on classification tasks; the representational dynamics in generative or self-supervised tasks might differ significantly.

Future Work Building on these results, we identify several promising directions:

- **Layer Merging**: Evaluating the performance impact of merging layers with high CKA similarity.
- **Residual Connections**: Investigating how skip-connections affect transformation capacity. While they propagate features (increasing similarity), they also allow for more complex transformations in the residual branch.
- **Transformer blocks**: Comparing the behavior of MLPs within Transformers to our standalone results, particularly the effect of the surrounding attention mechanism and normalization layers.

6 Conclusion

In this paper, we investigated the representational similarity across layers in deep Multi-Layer Perceptrons. Using Centered Kernel Analysis (CKA), we demonstrated that adjacent layers are inherently similar, confirming that individual layers perform incremental transformations. We quantified the "transformation capacity" of these layers and showed that it scales with layer width and is further utilized through the training process. Our findings suggest that the internal representations of MLPs are highly constrained by their architectural parameters, with wider layers allowing for more signif-

icant representational drift. These results provide new insights into the redundancy of deep models and offer a quantitative framework for future research into model compression and architectural optimization.

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